Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LEARNER GUIDE

For

Microsoft Azure AI Fundamentals (AI-900)

TGS Ref No: TGS-2023021100

Conducted by

Tertiary Infotech Academy Pte Ltd

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Learner Guide - Microsoft Azure AI Fundamentals (AI-900)

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Provider: Tertiary Infotech Academy Pte Ltd

Course Information

Item	Details
Course Title	Microsoft Azure AI Fundamentals (AI-900)
Course Code	TGS-2023021100
WSQ TSC	Artificial Intelligence Application (ICT-DIT-4016-1.1)
Duration	2 Days (16 training hours)
Delivery Mode	Instructor-led with guided Azure AI lab activities
Course Repository	https://github.com/tertiarycourses/TGS-2023021100---Microsoft-Azure-AI-Fundamentals-AI-900-

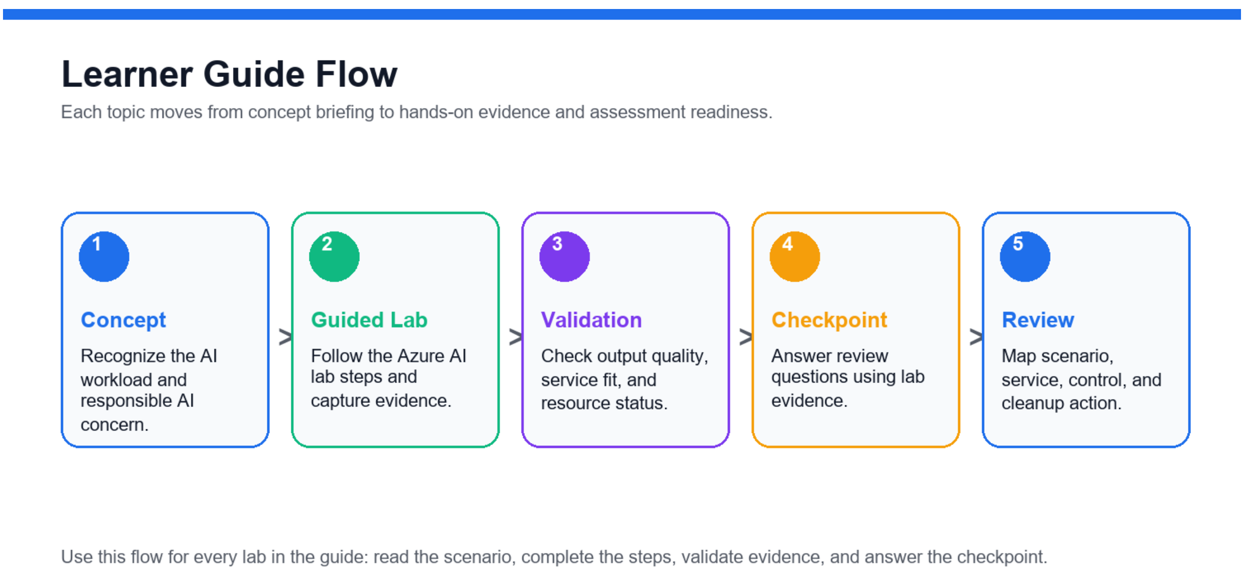
Certification Status Note

Microsoft Learn states that Exam AI-900 was retired on June 30, 2026. These materials are therefore written as Azure AI Fundamentals courseware aligned to the AI-900 skills outline and Azure AI concepts.

Course Goal

This course helps learners recognize common AI workloads, explain responsible AI considerations, and map business scenarios to Azure AI services through hands-on lab activities.

Course Learning Flow



Learning Outcomes

- Identify common AI workloads and use cases.
- Explain responsible AI principles and human review needs.
- Distinguish regression, classification, clustering, deep learning, and transformer-based workloads.
- Describe Azure Machine Learning and automated ML concepts.
- Map vision, NLP, speech, translation, document, and generative AI scenarios to Azure services.
- Recognize security, privacy, governance, and cost controls for AI solutions.

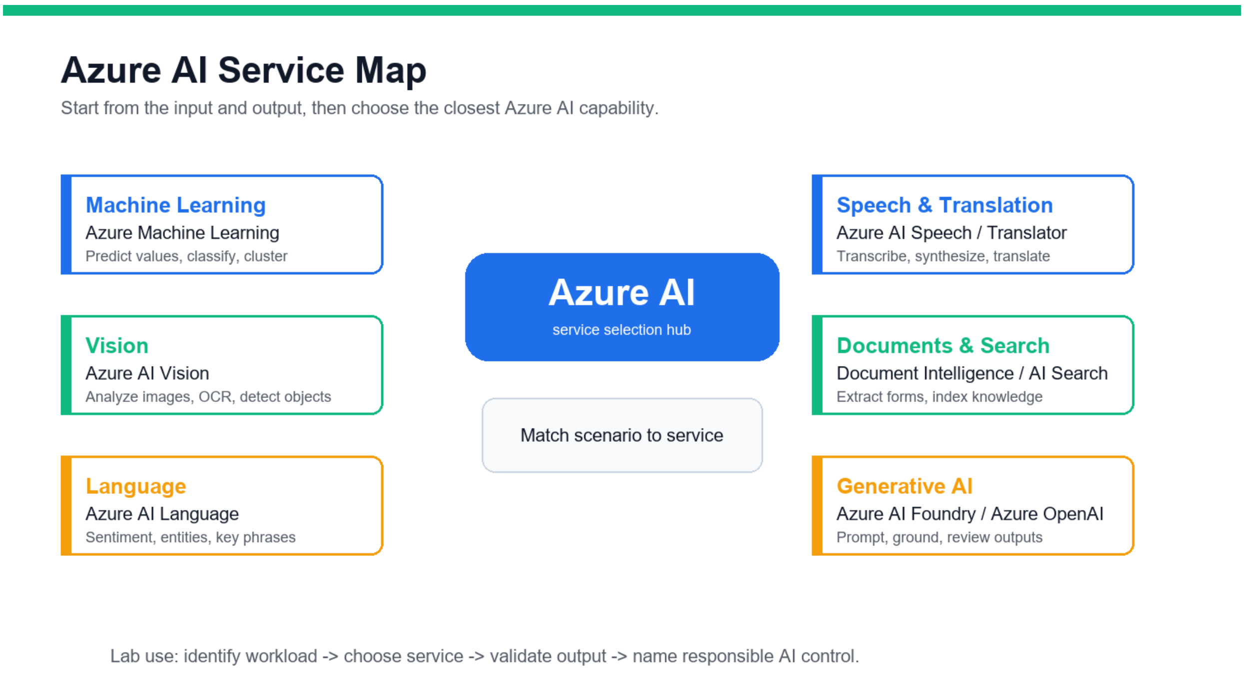
Skills Framework Alignment

TSC Title: Artificial Intelligence Application

TSC Code: ICT-DIT-4016-1.1

Knowledge / Ability	Lab Alignment
AI workload recognition and responsible AI	Labs 01, 09, 10
Machine learning concepts and Azure Machine Learning	Labs 02, 03
Vision, language, speech, and document AI services	Labs 04, 05, 06, 07
Generative AI, Azure AI Foundry, Azure OpenAI, and model catalog concepts	Lab 08
Governance, security, privacy, cost, and cleanup controls	Labs 09, 10

Azure AI Service Map



Lab Environment Setup

1. Open https://portal.azure.com/.
2. Sign in with an instructor-provided or personal Azure account.
3. Confirm subscription and resource group access.
4. Use free tiers where possible.
5. Do not paste personal or confidential data into AI demos.

6. Clean up resources when the trainer confirms it is safe.

Labs

Lab 01 - AI Workloads and Responsible AI

Objectives

- Identify common AI workloads.
- Match workloads to business scenarios.
- Explain responsible AI principles.
- Create a responsible AI checklist.

Scenario

A retail company wants to use AI for product recommendations, customer support, document processing, image analysis, and marketing content generation. You must identify the workload types and responsible AI concerns.

Steps

1. Identify AI Workloads

Create a table:

Scenario	AI Workload
Detect damaged products in photos	Computer vision
Extract invoice totals	Document processing
Classify customer feedback sentiment	Natural language processing
Generate campaign copy	Generative AI
Predict future sales	Machine learning

Add at least five more examples.

2. Map Workloads to Azure Services

Use this starting map:

Workload	Azure Service
Image analysis	Azure AI Vision
Text analytics	Azure AI Language
Speech to text	Azure AI Speech
Document extraction	Azure AI Document Intelligence
Generative AI	Azure AI Foundry or Azure OpenAI
Custom ML	Azure Machine Learning

3. Review Responsible AI Principles

Write a one-sentence explanation for:

```
Fairness
Reliability and safety
Privacy and security
Inclusiveness
Transparency
Accountability
```

4. Build a Responsible AI Checklist

For the retail use case, answer:

1. Who could be harmed by poor predictions?
2. What data is sensitive?
3. How will users know AI is involved?
4. Who approves the AI solution?
5. How are errors reported and corrected?

Validation

You should have workload examples, Azure service mapping, and a responsible AI checklist.

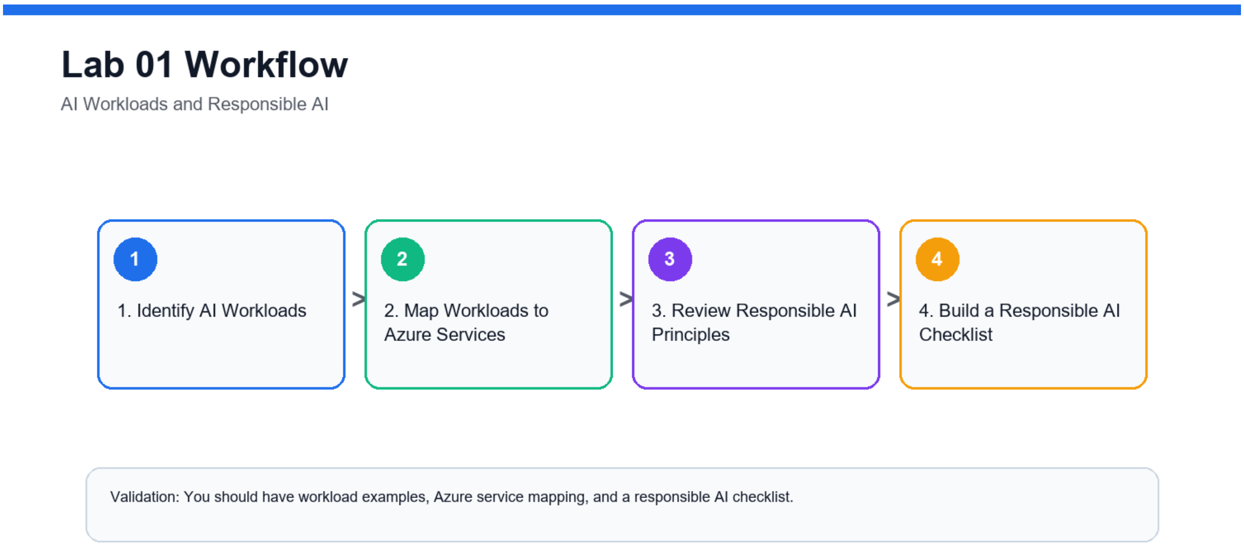
Checkpoint Questions

1. What is a computer vision workload?
2. Why is fairness important?
3. What is transparency in AI?
4. Who is accountable for an AI system?

Exam Focus

AI-900-style questions often ask you to identify workload types and responsible AI principles from a scenario.

Lab Workflow Diagram



Lab 02 - Machine Learning Fundamentals

Objectives

- Distinguish regression, classification, and clustering.
- Explain features, labels, training data, and validation data.
- Understand deep learning and transformer concepts.
- Match ML techniques to scenarios.

Scenario

The retail company wants to use historical data to predict outcomes. You must decide which machine learning technique fits each business question.

Steps

1. Classify ML Scenarios

Create a table:

Business Question	ML Technique
What will next month's sales be?	Regression
Will this customer churn?	Classification
Which customers behave similarly?	Clustering
Is this image a product or receipt?	Deep learning

2. Identify Features and Labels

For a churn prediction dataset, identify:

```
Features: customer age, tenure, purchase count, support tickets
Label: churned or not churned
```

Create two more examples.

3. Explain Training and Validation

Write:

1. Why data is split into training and validation sets.
2. What overfitting means.
3. Why evaluation metrics matter.

4. Review Deep Learning and Transformers

In your notes, explain:

- Deep learning uses neural networks with multiple layers.
- Transformers are useful for language and generative AI workloads.
- Foundation models can be adapted for many tasks.

5. Match Technique to Service

Need	Possible Azure Service
No-code ML model	Azure Machine Learning automated ML
Custom ML workflow	Azure Machine Learning
Generative text	Azure AI Foundry or Azure OpenAI
Image analysis without custom training	Azure AI Vision

Validation

You should have scenario mappings, feature/label examples, and a training/validation explanation.

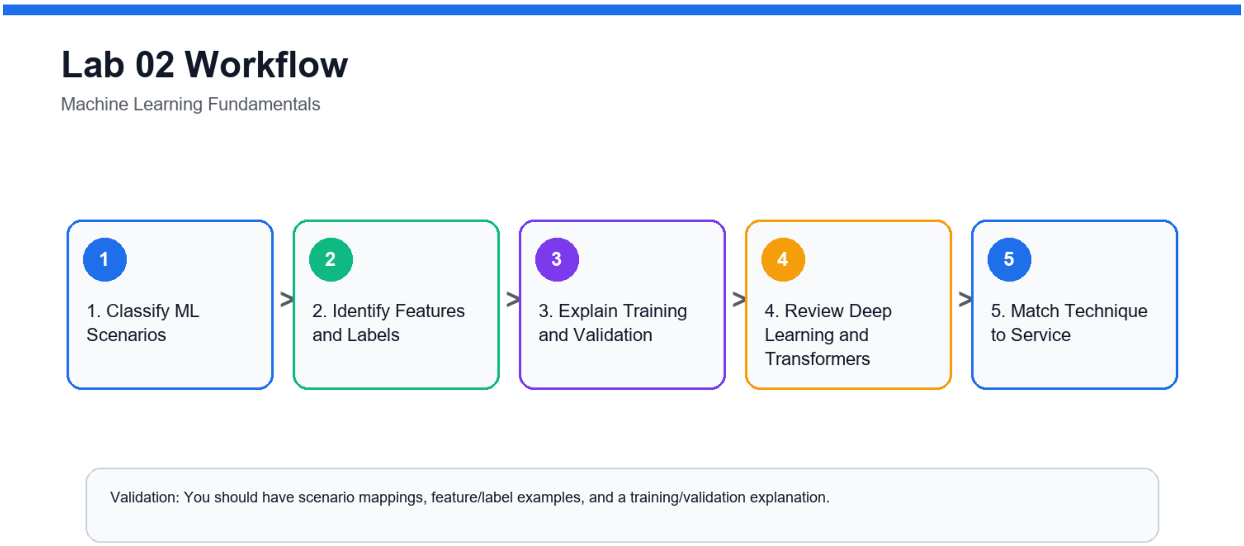
Checkpoint Questions

1. What is regression used for?
2. What is classification used for?
3. What is clustering used for?
4. What is the difference between a feature and a label?

Exam Focus

Know how to select the basic ML technique from the wording of a business scenario.

Lab Workflow Diagram



Lab 03 - Azure Machine Learning, Automated ML, Model Lifecycle

Objectives

- Describe Azure Machine Learning capabilities.
- Explain automated machine learning.
- Understand data, compute, jobs, models, and endpoints.
- Map the model lifecycle.

Scenario

The retail company wants to build a custom churn model but does not yet have a large data science team. You must explain how Azure Machine Learning can support the workflow.

Steps

1. Open Azure Machine Learning Studio

1. Open the Azure portal.
2. Search for Azure Machine Learning.
3. Open or create a training workspace if your instructor provides access.
4. Open Azure Machine Learning studio.

2. Identify Core Workspace Areas

Find and record:

```
Data
Notebooks
Compute
Jobs
Models
Endpoints
Automated ML
```

3. Explain Automated ML

Write how AutoML helps with:

- Trying multiple algorithms.
- Selecting features.
- Evaluating models.
- Ranking model candidates.
- Producing a model for deployment review.

4. Map the Model Lifecycle

Create a flow:

```
Collect data
Prepare data
Train model
Evaluate model
Register model
Deploy model
Monitor model
Retrain when needed
```

5. Choose Compute Safely

Discuss why training compute should match the lab size, budget, and workload.

Validation

You should have a workspace area map, AutoML explanation, and model lifecycle flow.

Cleanup

Delete any Azure Machine Learning workspace or compute resources created only for class if instructed.

Checkpoint Questions

1. What is automated machine learning?
2. What is a model endpoint?
3. Why is model monitoring needed?
4. Why does compute choice affect cost?

Exam Focus

Understand what Azure Machine Learning does at a high level: data, compute, training, automated ML, model management, deployment, and monitoring.

Lab Workflow Diagram

Lab 03 Workflow

Azure Machine Learning, Automated ML, Model Lifecycle

1. Open Azure Machine Learning Studio > 2. Identify Core Workspace Areas > 3. Explain Automated ML > 4. Map the Model Lifecycle > 5. Choose Compute Safely

Validation: You should have a workspace area map, AutoML explanation, and model lifecycle flow.

Lab 04 - Computer Vision, Image Analysis, OCR

Objectives

- Identify computer vision workloads.
- Distinguish image classification, object detection, OCR, and face detection.
- Describe Azure AI Vision capabilities.
- Review responsible AI concerns for vision.

Scenario

The retail company wants to analyze product photos, detect damaged items, read shelf labels, and blur faces in store images.

Steps

1. Match Vision Scenarios

Scenario	Vision Workload
Identify product category in an image	Image classification
Locate multiple products in a photo	Object detection
Read text from a receipt	Optical character recognition
Detect whether a face appears	Face detection

2. Explore Azure AI Vision Concepts

In the Azure portal or Microsoft Learn documentation, review:

```
Image analysis
OCR
Object detection
Image tagging
Caption generation
Face detection
```

3. Design a Vision Solution

For damaged product detection, document:

1. Input image source.

2. Expected output.

3. Azure service.

4. Human review requirement.

5. Responsible AI risk.

4. Responsible AI Review

Answer:

- Could image data contain people?
- Is consent required?
- Should images be retained?
- Could lighting or camera angle affect accuracy?
- How should users appeal incorrect results?

Validation

You should have a scenario mapping table, Azure AI Vision capability notes, and responsible AI review.

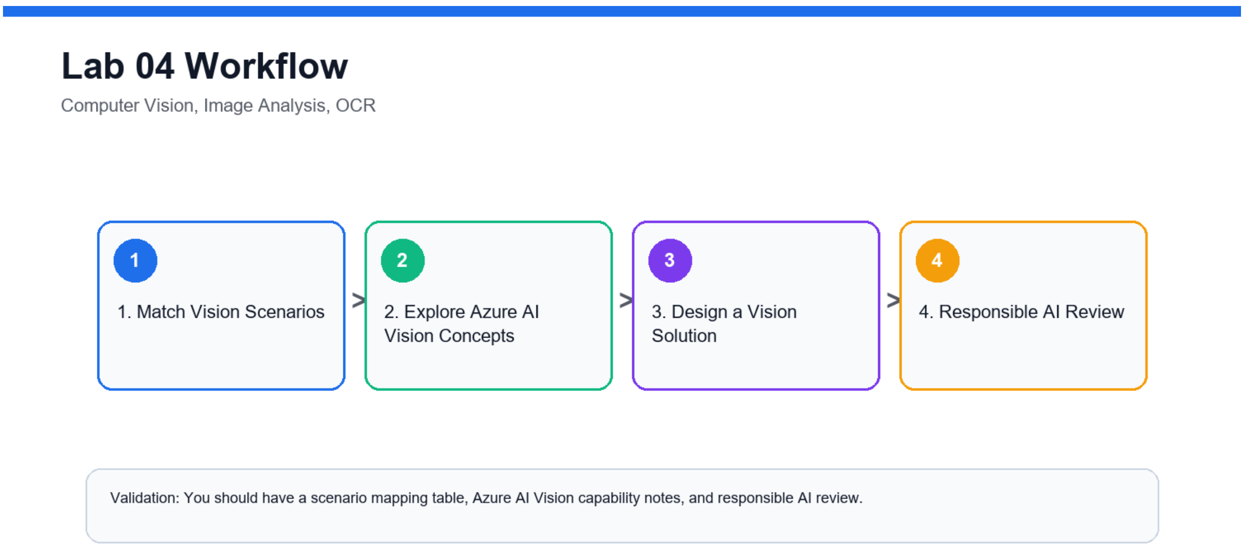
Checkpoint Questions

1. What is OCR?
2. How is object detection different from image classification?
3. Why is face analysis sensitive?
4. Which Azure service supports common vision workloads?

Exam Focus

Vision questions often ask which workload or Azure service fits the scenario.

Lab Workflow Diagram



Lab 05 - Natural Language Processing and Azure AI Language

Objectives

- Identify NLP workloads.
- Explain sentiment analysis, key phrase extraction, entity recognition, and language modeling.
- Describe Azure AI Language capabilities.
- Review responsible AI concerns for text processing.

Scenario

The retail company receives thousands of customer comments. It wants to detect sentiment, extract key topics, identify product names, and route complaints.

Steps

1. Map NLP Workloads

Scenario	NLP Workload
Is feedback positive or negative?	Sentiment analysis
What topics are mentioned?	Key phrase extraction
Which products or locations are named?	Entity recognition
What language is this text written in?	Language detection
What does this customer want?	Intent recognition or classification

2. Review Azure AI Language

Document capabilities such as:

```
Sentiment analysis
Key phrase extraction
Named entity recognition
Language detection
Text classification
Question answering
Summarization concepts
```

3. Design a Feedback Pipeline

Create a flow:

```
Collect feedback
Detect language
Translate if needed
Analyze sentiment
Extract key phrases
Identify entities
Route high-risk complaints
Store dashboard results
```

4. Responsible AI Review

Answer:

1. Could customer comments contain personal data?
2. Could sentiment analysis misunderstand sarcasm?
3. Should automated routing be reviewed by humans?
4. How should low-confidence results be handled?

Validation

You should have NLP workload mapping, Azure AI Language notes, and a feedback pipeline design.

Checkpoint Questions

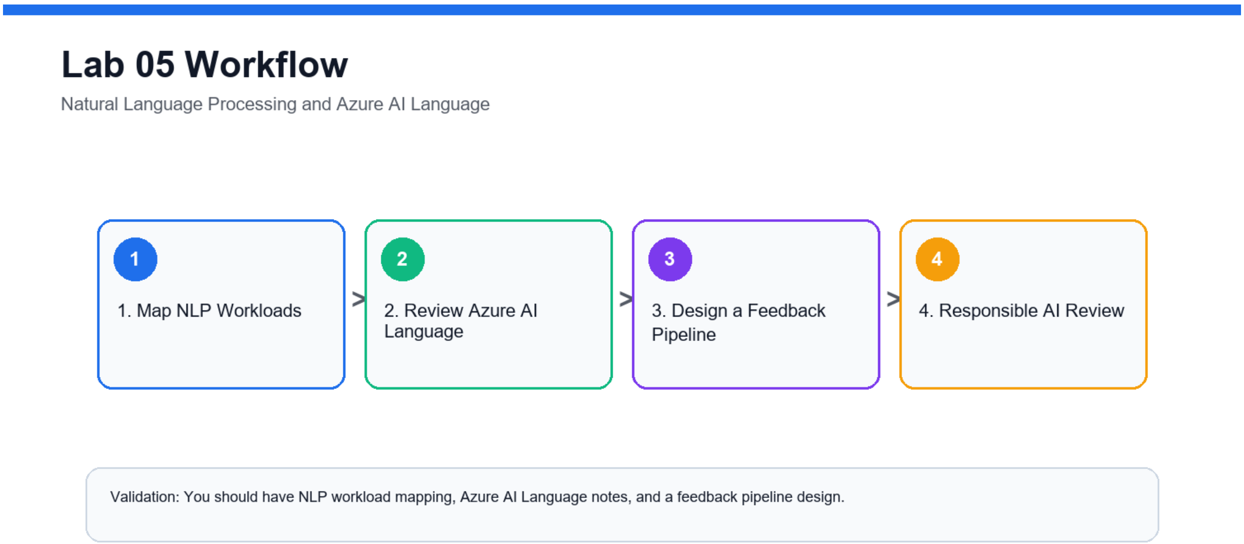
1. What is sentiment analysis?

2. What is entity recognition?
3. What is key phrase extraction?
4. Why can NLP results require human review?

Exam Focus

NLP questions often describe text, speech, translation, sentiment, or entity extraction scenarios.

Lab Workflow Diagram



Lab 06 - Speech, Translation, Conversational AI

Objectives

- Describe speech recognition and synthesis.
- Explain translation workloads.
- Recognize conversational AI use cases.
- Map scenarios to Azure AI Speech, Translator, and Bot-related services.

Scenario

The retail company wants a multilingual customer support experience with voice input, translated text, and a chatbot for common questions.

Steps

1. Match Speech and Translation Scenarios

Scenario	Workload
Convert customer voice to text	Speech recognition
Read a response aloud	Speech synthesis
Translate English to Malay	Translation
Detect spoken language	Speech or language detection
Answer FAQ through chat	Conversational AI

2. Review Azure Services

Document:

```
Azure AI Speech
Azure AI Translator
Azure Bot Service concepts
Azure AI Language for question answering
```

3. Design a Support Assistant

Create a flow:

```
Customer speaks
Speech to text
Language detection
Translation if required
Question answering or intent detection
Response generation
Text to speech
Escalate to human agent if needed
```

4. Responsible AI Review

Answer:

- What happens if speech recognition is wrong?
- How are accents and background noise handled?
- When should a human agent take over?
- Should users be told they are interacting with a bot?

Validation

You should have scenario mappings, Azure service notes, support assistant flow, and responsible AI review.

Checkpoint Questions

1. What is speech recognition?
2. What is speech synthesis?
3. What is translation used for?
4. Why is human escalation important in conversational AI?

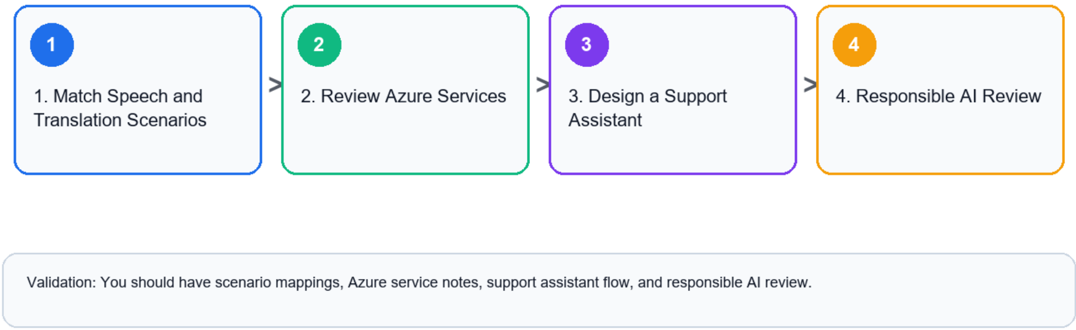
Exam Focus

Speech and translation questions usually test whether you can identify the workload and the right Azure AI service.

Lab Workflow Diagram

Lab 06 Workflow

Speech, Translation, Conversational AI



Lab 07 - Document Intelligence and Knowledge Mining

Objectives

- Identify document processing workloads.
- Explain form extraction and OCR.
- Describe Azure AI Document Intelligence.
- Explain knowledge mining concepts.

Scenario

The retail company processes invoices, receipts, and supplier forms manually. It wants to extract structured information and make documents searchable.

Steps

1. Identify Document Tasks

Document Task	Workload
Read printed text	OCR
Extract invoice number and amount	Form extraction
Extract tables	Document intelligence
Search across scanned files	Knowledge mining
Classify document type	Classification

2. Review Azure AI Document Intelligence

Document:

```
Prebuilt models
Custom extraction
Layout extraction
Tables
Key-value pairs
Confidence scores
```

3. Design an Invoice Processing Flow

```
Upload invoice
```

```
Run document extraction
Validate confidence score
Human review if low confidence
Store structured fields
Search and report
Archive according to policy
```

4. Review Knowledge Mining

Explain how AI enrichment, indexing, and search can make unstructured documents discoverable.

5. Responsible AI Review

Answer:

- Do documents contain personal data?
- What confidence threshold triggers human review?
- Who can access extracted data?
- How long are documents retained?

Validation

You should have a document task map, extraction flow, knowledge mining notes, and responsible AI review.

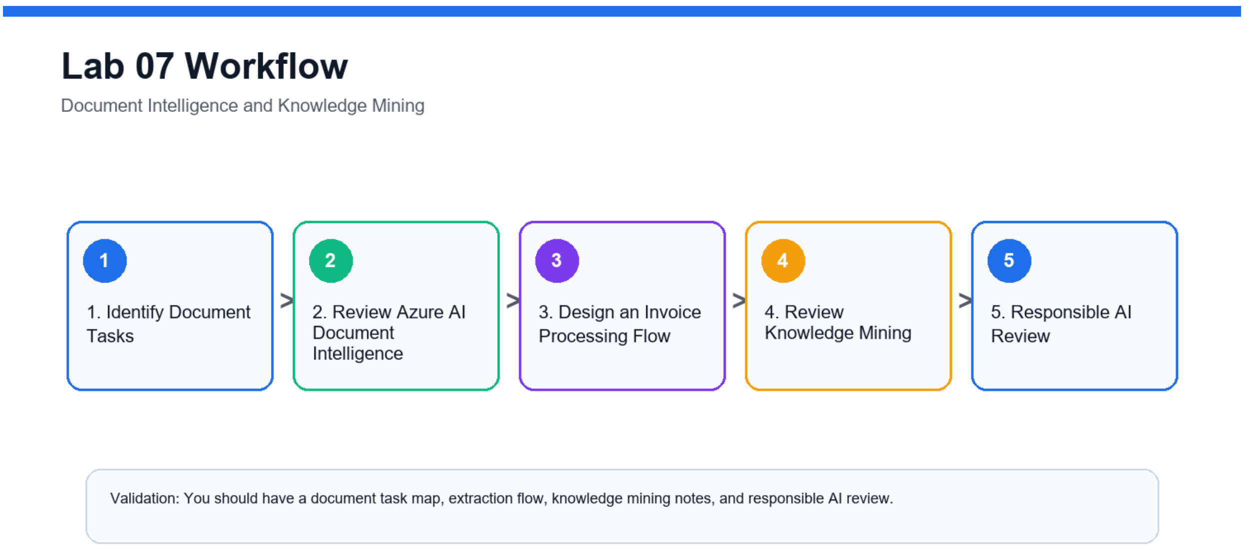
Checkpoint Questions

1. What is document processing?
2. How is OCR related to document intelligence?
3. Why are confidence scores useful?
4. What is knowledge mining?

Exam Focus

Document processing questions may reference OCR, forms, invoices, extraction, and search.

Lab Workflow Diagram



Lab 08 - Generative AI, Azure AI Foundry, Azure OpenAI, Model Catalog

Objectives

- Explain generative AI concepts.
- Identify common generative AI scenarios.
- Describe Azure AI Foundry, Azure OpenAI, and model catalog capabilities.
- Apply responsible AI considerations for generative AI.

Scenario

The retail company wants to use generative AI to draft product descriptions, summarize customer reviews, and help support agents respond faster.

Steps

1. Define Generative AI Terms

Create glossary entries:

```
Generative AI
Prompt
Completion
Foundation model
Large language model
Transformer
Grounding
Retrieval augmented generation
Content filter
```

2. Match Generative AI Scenarios

Scenario	Generative AI Use
Draft product description	Text generation
Summarize reviews	Summarization
Answer questions from policy docs	Grounded question answering
Create support response draft	Agent assistance
Generate image concepts	Image generation

3. Review Azure Capabilities

Document:

```
Azure AI Foundry
Azure OpenAI service
Model catalog
Prompt flow concepts
Content safety
Evaluation
```

4. Responsible AI Review

Answer:

1. Could the model produce inaccurate content?
2. Does output need human review?
3. What data should not be sent to the model?
4. How are harmful outputs filtered?
5. How will users know content is AI-assisted?

5. Design a Safe Assistant

Create a short design:

```
Use case:
Data source:
```

```
Model:
Grounding approach:
Human review:
Monitoring:
Safety controls:
```

Validation

You should have a generative AI glossary, scenario map, Azure capability notes, and safe assistant design.

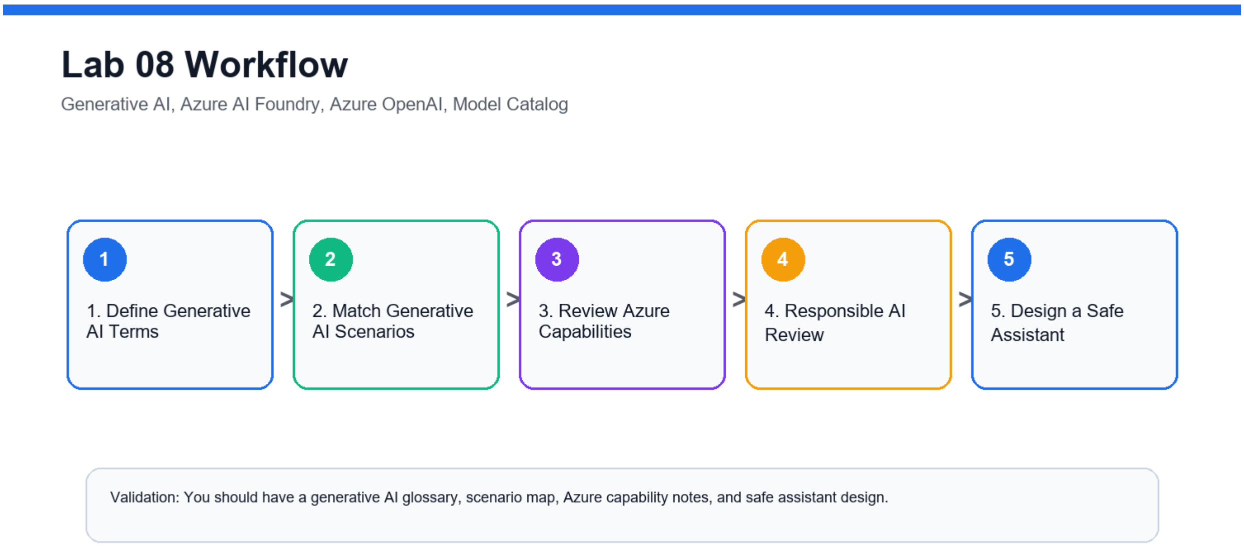
Checkpoint Questions

1. What is generative AI?
2. What is a prompt?
3. What is grounding?
4. Why is responsible AI especially important for generative AI?

Exam Focus

Current AI fundamentals material emphasizes generative AI, Azure AI Foundry, Azure OpenAI, and model catalog concepts.

Lab Workflow Diagram



Lab 09 - AI Solution Design, Security, Cost, Governance

Objectives

- Design a simple Azure AI solution.
- Identify security, privacy, and governance controls.
- Review cost considerations.
- Choose appropriate Azure AI services.

Scenario

The retail company wants one AI solution that analyzes customer feedback, summarizes themes, and routes urgent complaints. You must recommend services and controls.

Steps

1. Confirm Requirements

Write:

```
Business goal:
Input data:
Expected output:
Users:
Latency need:
Privacy constraints:
Human review requirement:
```

2. Choose Services

Requirement	Azure Service
Sentiment and entities	Azure AI Language
Translation	Azure AI Translator
Summarization or response drafting	Azure AI Foundry or Azure OpenAI
Custom ML prediction	Azure Machine Learning
Secure storage	Azure Storage with access control

3. Add Security Controls

Document:

- Role-based access control.
- Managed identities.
- Private data handling.
- Logging and monitoring.
- Key and endpoint protection.
- Human review for high-risk actions.

4. Review Cost Controls

Answer:

1. Which services are billed per transaction or token?
2. How will usage be monitored?
3. What budget alert is needed?
4. How will unused resources be deleted?

5. Add Responsible AI Controls

Include:

- Transparency message.
- Error reporting channel.
- Human review workflow.
- Bias and fairness review.
- Privacy review.

Validation

You should have a requirements summary, service selection table, security controls, cost controls, and responsible AI controls.

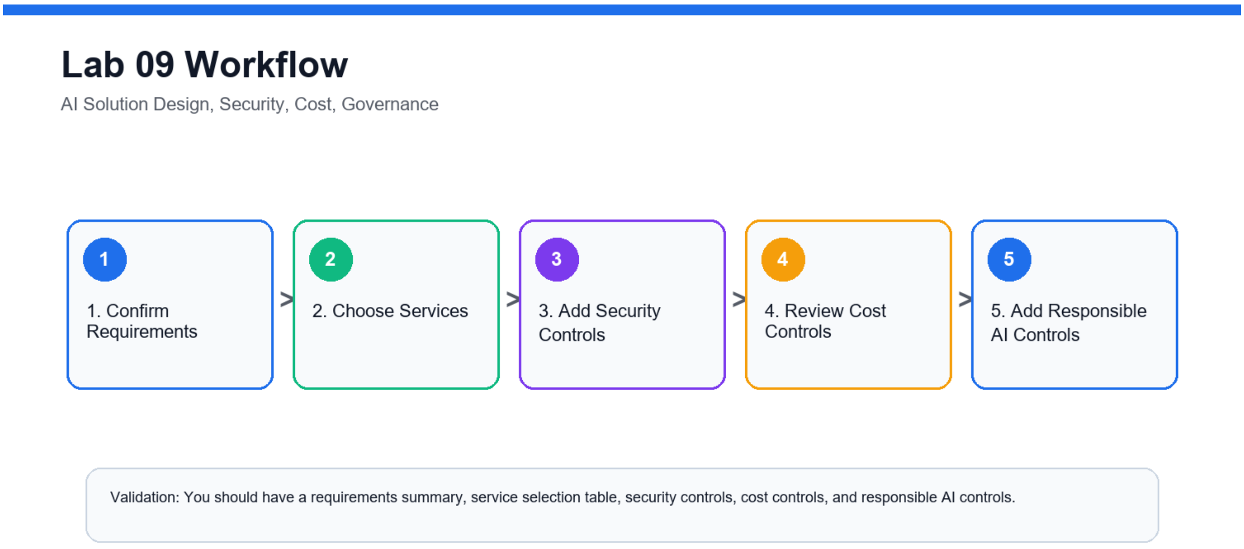
Checkpoint Questions

1. Why does AI solution design include governance?
2. What is RBAC used for?
3. Why are budget alerts useful?
4. Why is human review important for high-impact outputs?

Exam Focus

AI solution questions often test choosing the right service and recognizing responsible AI, privacy, and security concerns.

Lab Workflow Diagram



Lab 10 - AI-900 Capstone and Course Review

Objectives

- Review all Azure AI fundamentals topics.
- Build an end-to-end AI service map.
- Practice workload recognition.
- Create a personal review plan.
- Clean up lab resources.

Scenario

You must brief a manager on which Azure AI services could support the retail company and what risks must be managed.

Steps

1. Build a Service Map

Create a table:

| Business Need | AI Workload | Azure Service |

| --- | --- | --- |

| Analyze product photos | Computer vision | Azure AI Vision |

| Extract invoice fields | Document processing | Azure AI Document Intelligence |

| Understand customer comments | NLP | Azure AI Language |

| Convert calls to text | Speech recognition | Azure AI Speech |

| Draft response suggestions | Generative AI | Azure AI Foundry or Azure OpenAI |

| Train custom prediction model | Machine learning | Azure Machine Learning |

2. Review Responsible AI

For each principle, write one action:

```
Fairness:
Reliability and safety:
Privacy and security:
Inclusiveness:
Transparency:
Accountability:
```

3. Review ML Techniques

Match:

| Technique | Scenario |

| --- | --- |

| Regression | Predict numeric value |

| Classification | Predict category |

| Clustering | Group similar items |

| Deep learning | Complex image, speech, or language tasks |

| Transformer | Language and generative AI workloads |

4. Build a Review Log

Record weak areas:

```
Topic:
Why it is confusing:
Definition to memorize:
Example scenario:
```

5. Clean Up Resources

If resources were created:

```
az group delete --name rg-ai900-lab --yes --no-wait
```

Confirm with your instructor before deleting shared resources.

6. Prepare a 7-Day Study Plan

1. Day 1: AI workloads and responsible AI.
2. Day 2: Machine learning fundamentals.
3. Day 3: Azure Machine Learning.
4. Day 4: Computer vision and document intelligence.
5. Day 5: NLP, speech, and translation.
6. Day 6: Generative AI, Foundry, Azure OpenAI.
7. Day 7: Scenario practice and review log.

Validation

You should have a service map, responsible AI action list, ML technique table, review log, and cleanup confirmation.

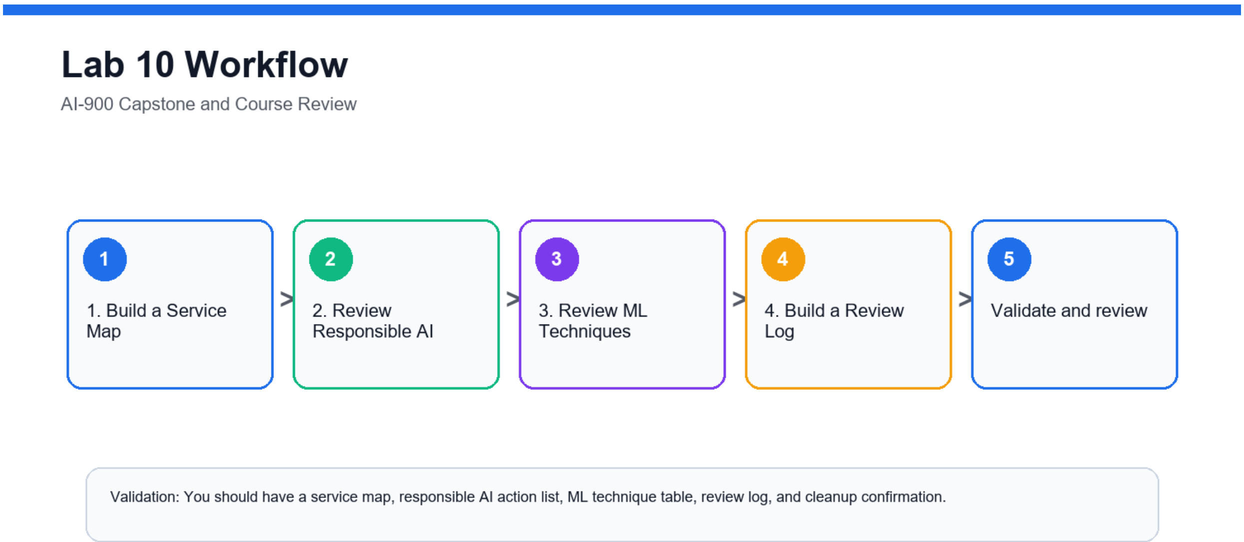
Checkpoint Questions

1. Which AI workload is easiest for you to identify?
2. Which Azure service names are most confusing?
3. What is the difference between traditional ML and generative AI?
4. What responsible AI principle do you need to review most?

Course Focus

The goal is to recognize AI workloads, understand foundational concepts, and describe how Azure AI services support common business scenarios.

Lab Workflow Diagram



Final Review Checklist

- Match AI workloads to Azure services.
- Explain responsible AI principles.
- Distinguish machine learning techniques.
- Recognize vision, language, speech, document, and generative AI scenarios.
- Identify security, privacy, cost, and governance considerations.